

EcoBuck Smart Compost Monitoring System

6-Week Connectivity & Backend Execution Roadmap (Assignment C)

Author: Toqeer Ahmed

Role: Principal IoT Systems & Backend Architect

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Status: PROPOSED (Roadmap Review Stage)

Week 1: Requirements Locking, Cloud Path Selection & Baseline Project Init

Day 1: Project Setup and Local Environment Provisioning

- **Objective:** Set up python environment, dependency management tools, local Git configurations, and initial repository scaffolding.
- **Expected Output:** Functioning local environment running Python 3.10+, virtual environment active, and baseline dependencies installed.
- **Dependencies:** None.
- **Skills Required:** Git version control, Python package management (`pip`, `venv`).
- **Resources to Learn:** [Python virtual environments guide](#).
- **Deliverables:** Main project directory structure with `.gitignore`, `requirements.txt`, and virtual environment directory.
- **Potential Risks:** Local dependency version mismatches with other team members.
- **Acceptance Criteria:** A command `python -m pip list` runs successfully and version control tracks initial structure.
- **Collaboration Required:** Aligning Git branching guidelines with Firmware (Team B) and Mobile (Team D) leads.

Day 2: Cloud Platform Comparative Analysis & Architecture Selection

- **Objective:** Document and decide between PostgreSQL/TimescaleDB, Firestore NoSQL, and MQTT setups.
- **Expected Output:** Signed-off Architecture Decision Record (ADR-001) outlining final database choice (Firestore).
- **Dependencies:** Day 1 local environment configuration.
- **Skills Required:** Technology stack evaluation, cost estimation, system design analysis.
- **Resources to Learn:** [Firestore pricing and performance limits](#).
- **Deliverables:** `docs/adr/001-cloud-database-selection.md` in repository.
- **Potential Risks:** Picking a database structure that is too complex for 6-week prototyping.
- **Acceptance Criteria:** The ADR is written, detailing costs, query speeds, complexity tradeoffs, and mobile integration speeds.
- **Collaboration Required:** Approval from Product Lead and Assignment D (Mobile App) owner.

Day 3: Finalizing Ingestion Schema Contracts

- **Objective:** Define the exact telemetry JSON payload schema that the ESP32 firmware will send.
- **Expected Output:** API contract document specifying exact variable names, types, units, and boundaries.
- **Dependencies:** Day 2 database choice selection.
- **Skills Required:** JSON schema layout, data contracting.
- **Resources to Learn:** [JSON Schema standards](#).
- **Deliverables:** `docs/schemas/telemetry_schema.json` documenting the JSON contract.
- **Potential Risks:** Firmware team sending fields not supported by the backend parsing models.
- **Acceptance Criteria:** Signed-off schema definition containing `device_id`, timestamp, readings (raw/filtered/trend), status, quality flags, and battery level.
- **Collaboration Required:** Joint session with Team B (Firmware) to freeze key names.

Day 4: Core Project Structure Scaffolding

- **Objective:** Implement folder structures matching the clean architecture design (app, api, services, schemas, repositories, core).
- **Expected Output:** Empty Python files initialized with imports and structure templates.
- **Dependencies:** Day 1 & Day 3 contract freezes.
- **Skills Required:** FastAPI structure layouts, clean architecture paradigms.
- **Resources to Learn:** [FastAPI Bigger Applications Project Layout](#).

- **Deliverables:** Folders `/app`, `/app/api`, `/app/schemas`, `/app/services`, `/app/repositories`, `/app/db`.
- **Potential Risks:** Circular imports due to poor scaffolding configurations.
- **Acceptance Criteria:** Command `uvicorn app.main:app` successfully runs an empty API server on port 8000 without warnings.
- **Collaboration Required:** Internal alignment only.

Day 5: Milestone 1 Review: Scope Signoff and ADR Locking

- **Objective:** Review scope, check constraints, and lock requirements for Milestone 1 (M1).
- **Expected Output:** Signed milestone agreement and updated project board backlog.
- **Dependencies:** Days 1-4 deliverables completed.
- **Skills Required:** Project management, task planning.
- **Resources to Learn:** [Agile sprint retrospective formats](#).
- **Deliverables:** Milestone 1 evidence log.
- **Potential Risks:** Undetected scope creep stretching the 6-week timeframe.
- **Acceptance Criteria:** Product lead approves scope notes and decision log.
- **Collaboration Required:** All Assignment owners (A, B, C, D) and Product Lead must sign off.

Week 2: Authentication & Ingestion Baseline Prototype

Day 6: Device Authentication System Implementation

- **Objective:** Build token validation logic to secure the device ingestion path.
- **Expected Output:** Core security helper verifying API keys from `X-Device-Token` header.
- **Dependencies:** Core folder scaffolding.
- **Skills Required:** Symmetric encryption, hashing algorithms, API design.
- **Resources to Learn:** [FastAPI Security Helpers](#).
- **Deliverables:** `app/core/security.py` containing token generation and hash verification helper scripts.
- **Potential Risks:** Hardcoding secrets in git repo histories.
- **Acceptance Criteria:** Passing security test checks showing valid tokens bypass routing but invalid tokens fail.

- **Collaboration Required:** Alignment with Team B to coordinate how tokens are stored in the device's flash memory.

Day 7: Mobile Authentication Setup

- **Objective:** Set up standard user OAuth2 token exchanges using JWTs.
- **Expected Output:** Endpoint router generating JWT access tokens from email/password verify.
- **Dependencies:** Day 6 security helpers.
- **Skills Required:** JSON Web Tokens (JWT), password hashing (`passlib`, `bcrypt`).
- **Resources to Learn:** [OAuth2 with Password and Bearer](#).
- **Deliverables:** `/app/api/v1/endpoints/auth.py` registering endpoints and `/app/schemas/auth.py` validation rules.
- **Potential Risks:** Using weak signing keys or insecure validation mechanisms.
- **Acceptance Criteria:** Postman requests to `/api/v1/auth/login` successfully yield JWT strings expiring in 60 minutes.
- **Collaboration Required:** Assignment D (Mobile App) team to align JWT handling protocols.

Day 8: Ingestion Endpoint Prototype Creation

- **Objective:** Program the initial ingestion route `/api/v1/telemetry`.
- **Expected Output:** API handler receiving incoming JSON body and print logging contents.
- **Dependencies:** Day 3 schema lock, Day 6 device auth.
- **Skills Required:** FastAPI router methods, request body parsing.
- **Resources to Learn:** [FastAPI path parameters and body payloads](#).
- **Deliverables:** `/app/api/v1/endpoints/telemetry.py` containing the POST endpoint.
- **Potential Risks:** Processing payloads with incorrect Content-Types.
- **Acceptance Criteria:** An HTTP client sending valid telemetry payload receives HTTP 202 Accepted.
- **Collaboration Required:** Internal developer tests.

Day 9: Payload Validation Logic Mapping

- **Objective:** Bind the telemetry Pydantic schemas to the ingestion route, validating inputs at runtime.
- **Expected Output:** Schema filters checking payload parameters (timestamps, float boundaries, array lengths).
- **Dependencies:** Day 8 endpoint prototype.

- **Skills Required:** Pydantic validation decorators, constraint bindings.
- **Resources to Learn:** [Pydantic V2 validation types](#).
- **Deliverables:** `/app/schemas/telemetry.py` mapping structure and properties.
- **Potential Risks:** Over-validating values causing normal sensor fluctuations to get rejected.
- **Acceptance Criteria:** Telemetry posts containing missing fields yield clear validation error structures (HTTP 422).
- **Collaboration Required:** Review with Team B to ensure firmware limits match validation thresholds.

Day 10: Milestone 2 Review: Local Mock Ingestion Demo

- **Objective:** Present a working demonstration of Milestone 2 (M2) core ingestion function.
- **Expected Output:** Terminal demonstrations showcasing telemetry ingest pipelines.
- **Dependencies:** Days 6-9 deliverables.
- **Skills Required:** Technical presentation, testing pipelines.
- **Resources to Learn:** [Designing software demos](#).
- **Deliverables:** Ingestion prototype demonstration logs.
- **Potential Risks:** Network configuration errors during demonstration.
- **Acceptance Criteria:** Demonstration runs without errors, showing a mock payload successfully authenticated and validated.
- **Collaboration Required:** Relevant owners (B, D) and Product Lead must attend.

Week 3: Firestore/PostgreSQL Database Schema & Caching Layer

Day 11: Database Registry & CRUD Scaffolding

- **Objective:** Setup database clients and base CRUD repositories.
- **Expected Output:** Singleton database connections initialized on server startup.
- **Dependencies:** Cloud configuration parameters.
- **Skills Required:** Firebase SDK configuration, NoSQL clients.
- **Resources to Learn:** [Firebase Admin SDK for Python setup](#).
- **Deliverables:** `/app/db/firestore.py` database connection file and `/app/repositories/base.py` CRUD interfaces.

- **Potential Risks:** Connection timeouts or improper client initializations.
- **Acceptance Criteria:** The application connects to database successfully without throwing credential errors.
- **Collaboration Required:** Product Lead to provision cloud credentials.

Day 12: Time-Series Telemetry Persistence Implementation

- **Objective:** Write incoming telemetry data points into the telemetry storage.
- **Expected Output:** Telemetry payloads written to database collections.
- **Dependencies:** Day 11 database setup, Day 9 validation logic.
- **Skills Required:** Database writes, document layouts.
- **Resources to Learn:** [Firestore write operations](#).
- **Deliverables:** `/app/repositories/telemetry.py` repository class.
- **Potential Risks:** High storage utilization due to unnecessary fields (e.g., redundant text strings).
- **Acceptance Criteria:** Database records are created containing exact timestamp mappings and sensor values.
- **Collaboration Required:** Internal tests.

Day 13: Caching Latest Status

- **Objective:** Implement latest status write-duplication caching pattern.
- **Expected Output:** Ingestion pipeline updates single status document upon writing time-series log.
- **Dependencies:** Day 12 telemetry repository.
- **Skills Required:** Cache strategy design, database transaction executions.
- **Resources to Learn:** [Firestore Transactions and Batched Writes](#).
- **Deliverables:** Add latest status updating functions inside `/app/repositories/device.py`.
- **Potential Risks:** Race conditions causing latest status to desynchronize from time-series logs.
- **Acceptance Criteria:** Every telemetry write results in updating `/devices/{id}/latest/status` document with current state.
- **Collaboration Required:** Align details with Mobile App Team D.

Day 14: Historical Queries & Pagination

- **Objective:** Create historical data fetching routines with limits and ordering filters.
- **Expected Output:** Retrieve paginated arrays of telemetry points sorted by timestamp.

- **Dependencies:** Day 12 telemetry database storage.
- **Skills Required:** NoSQL query construction, date filters.
- **Resources to Learn:** [Querying data in Firestore](#).
- **Deliverables:** Query filters in `/app/repositories/telemetry.py`.
- **Potential Risks:** High latency queries if indexes are missing.
- **Acceptance Criteria:** Running the queries returns data points ordered by timestamp within requested window.
- **Collaboration Required:** Mobile Team D to design chart API pagination parameters.

Day 15: Database Access Rules and Indexing Rules Setup

- **Objective:** Secure collections and build database indexes for ordering constraints.
- **Expected Output:** Security configuration rules deployed to cloud console.
- **Dependencies:** Database collections setup.
- **Skills Required:** Database access control, index provision.
- **Resources to Learn:** [Firestore Security Rules syntax](#).
- **Deliverables:** `firestore.rules` file containing security specifications.
- **Potential Risks:** Over-restrictive database rules blocking legitimate requests.
- **Acceptance Criteria:** Unauthenticated reads to user data are rejected; search queries requiring compound sort run without index errors.
- **Collaboration Required:** Security lead review if applicable.

Week 4: Alert Rules Engine & Notifications Hooking

Day 16: Alert Engine Core Logic Setup

- **Objective:** Scaffold the rules evaluator processing data during ingestion.
- **Expected Output:** Execution pipeline routing telemetry data into checking steps.
- **Dependencies:** Week 3 database storage.
- **Skills Required:** Event processing systems, modular programming.
- **Resources to Learn:** [Designing rules engines](#).
- **Deliverables:** `/app/alert_engine/processor.py` base script.
- **Potential Risks:** Lag in ingestion pipelines if alert analysis is slow.

- **Acceptance Criteria:** Processing test payload invokes the check routines.
- **Collaboration Required:** Internal tests.

Day 17: Rule Boundaries Implementation

- **Objective:** Program logic checks (high heat threshold, extreme moisture limits, error flags).
- **Expected Output:** Functions evaluating telemetry values and flagging true/false states.
- **Dependencies:** Day 16 engine core.
- **Skills Required:** Conditional evaluations, domain math.
- **Resources to Learn:** [Composting guidelines for temperature and moisture limits](#).
- **Deliverables:** `/app/alert_engine/rules.py` file with bounds evaluations.
- **Potential Risks:** High rate of false positives on noisy inputs.
- **Acceptance Criteria:** Readings with moisture < 20% or temp > 65°C successfully trigger flag changes.
- **Collaboration Required:** Check thresholds with Assignment A (Sensors/Calibration) team.

Day 18: Heartbeat/Offline Monitor Setup

- **Objective:** Create background scheduler to flag devices that stop reporting.
- **Expected Output:** A routine running checks on the last seen timestamp of all devices.
- **Dependencies:** Day 17 rules engine.
- **Skills Required:** Background task execution, cron jobs.
- **Resources to Learn:** [FastAPI Background Tasks](#).
- **Deliverables:** Heartbeat checking schedule in `/app/alert_engine/processor.py`.
- **Potential Risks:** Server resources saturated by constant scanning loops.
- **Acceptance Criteria:** Setting device timestamp back 20 minutes triggers an offline status and alerts.
- **Collaboration Required:** Internal developers testing.

Day 19: Firebase Cloud Messaging Integration

- **Objective:** Set up push notifications to forward alerts to mobile devices.
- **Expected Output:** FCM client connection dispatching push envelopes.
- **Dependencies:** Day 17 rules engine, Firebase project config.
- **Skills Required:** Push notification APIs, FCM integration.

- **Resources to Learn:** [FCM Python SDK documentation](#).
- **Deliverables:** `/app/services/notification.py` integrations.
- **Potential Risks:** Network blockages or device token invalidations.
- **Acceptance Criteria:** Calling dispatch helper yields success message from Google servers.
- **Collaboration Required:** Mobile Team D to register mobile device registration tokens.

Day 20: Milestone 3 Review: Integration Schema Freeze

- **Objective:** Freeze data schemas, status names, and routes for Milestone 3 (M3).
 - **Expected Output:** Complete API contracts frozen and published.
 - **Dependencies:** Days 11-19 completed.
 - **Skills Required:** Quality assurance, API freeze checkpoints.
 - **Resources to Learn:** [API Versioning best practices](#).
 - **Deliverables:** Versioned API schema contract log.
 - **Potential Risks:** Retrospective schema changes breaking build files.
 - **Acceptance Criteria:** Developers B, C, D sign agreement on frozen variables and routes.
 - **Collaboration Required:** All Assignment Owners (A, B, C, D) must attend.
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Week 5: Logging, System Monitoring & Mock Simulator

Day 21: Logging and Request Tracing Setup

- **Objective:** Create standardized logs for server actions and error states.
- **Expected Output:** Middleware tagging incoming API calls and writing structured logs.
- **Dependencies:** Core configuration utilities.
- **Skills Required:** Log formats, tracing, HTTP middlewares.
- **Resources to Learn:** [Python logging module](#).
- **Deliverables:** `/app/utils/logger.py` format configuration.
- **Potential Risks:** Verbose logs filling server disk spaces.
- **Acceptance Criteria:** Every HTTP call yields formatted logger output displaying elapsed time.
- **Collaboration Required:** Internal tests.

Day 22: Analytics and Metric Counters

- **Objective:** Collect system performance metrics (number of successful uploads, alert frequencies).
- **Expected Output:** Endpoint returning system metrics and counters.
- **Dependencies:** Week 3 database storage.
- **Skills Required:** Aggregate queries, dashboard metrics.
- **Resources to Learn:** [REST API analytics design](#).
- **Deliverables:** `/api/v1/endpoints/alerts.py` summary charts endpoints.
- **Potential Risks:** High database overhead for tracking metrics.
- **Acceptance Criteria:** Calling metric endpoint returns current upload counts and alert logs.
- **Collaboration Required:** Coordinate with Mobile App Team D.

Day 23: Complete Mock Device Simulator Script

- **Objective:** Write simulator script generating realistic sensor data curves.
- **Expected Output:** Python simulator utility pushing telemetry updates via HTTP POST requests.
- **Dependencies:** Ingestion routes deployed.
- **Skills Required:** Telemetry simulation, HTTP client tools (`requests`, `httplib`).
- **Resources to Learn:** [Building HTTP mock clients in Python](#).
- **Deliverables:** `/simulator/run_simulation.py` script.
- **Potential Risks:** Simulator flooding server if loop intervals fail.
- **Acceptance Criteria:** Simulator script runs, logging successful 202 responses from the ingestion endpoint.
- **Collaboration Required:** Deliver script to Mobile Team D to unblock client tests.

Day 24: Integration Test Suite Execution

- **Objective:** Run tests validating endpoints, database transactions, and rules evaluations.
- **Expected Output:** Pytest output showing all test cases successfully pass.
- **Dependencies:** All core modules deployed.
- **Skills Required:** Unit testing, mock techniques.
- **Resources to Learn:** [Testing FastAPI applications with Pytest](#).
- **Deliverables:** `/tests/` collection scripts.
- **Potential Risks:** Fragile tests generating false failures.

- **Acceptance Criteria:** Execution of `pytest` displays success code across all tests.
- **Collaboration Required:** Review test coverage with Quality Lead.

Day 25: Milestone 4 Review: Security Auditing & Bench Testing

- **Objective:** Review security and execute bench testing validation for Milestone 4 (M4).
 - **Expected Output:** Vulnerability logs and verified proof-of-concept tests.
 - **Dependencies:** Week 5 deliverables.
 - **Skills Required:** Security audit, hardware-in-the-loop validation.
 - **Resources to Learn:** [OWASP IoT Top 10 Security checks](#).
 - **Deliverables:** Milestone 4 test report.
 - **Potential Risks:** Security issues requiring redesign of authorization paths.
 - **Acceptance Criteria:** Zero critical issues, authentication rules pass validation.
 - **Collaboration Required:** Assignment B (Firmware) and A (Sensors) leads must participate.
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Week 6: Full End-to-End Integration, Handover & Demo Readiness

Day 26: Cross-Team Integration Testing

- **Objective:** Perform tests verifying hardware-to-cloud-to-mobile communications.
- **Expected Output:** Live data uploads from physical ESP32 boards displaying on the mobile app.
- **Dependencies:** Milestone 3 freezes, Mobile app integration.
- **Skills Required:** End-to-end integration testing, wireless networks debugging.
- **Resources to Learn:** [Cross-platform end-to-end testing systems](#).
- **Deliverables:** Integration verification log.
- **Potential Risks:** High latency or message loss on physical wireless networks.
- **Acceptance Criteria:** Physical device status change updates mobile screen in under 5 seconds.
- **Collaboration Required:** Joint session with Team B (Firmware) and Team D (Mobile App).

Day 27: Manual Validation Endpoint Verification

- **Objective:** Verify manual compost validation flows.
- **Expected Output:** Validation command from mobile transitions status inside database and synced back to device config.
- **Dependencies:** Day 26 integration.
- **Skills Required:** Transaction systems, configuration synchronization.
- **Resources to Learn:** [Design of commands and events in REST APIs](#).
- **Deliverables:** Integration checks for `/validate` route.
- **Potential Risks:** Race conditions during reset transitions.
- **Acceptance Criteria:** Triggering validation transitions state to `ready_confirmed` and changes status on physical device during next sync.
- **Collaboration Required:** Teams B and D.

Day 28: Deployment Packaging

- **Objective:** Package application inside Docker containers.
- **Expected Output:** Built docker image running FastAPI service locally.
- **Dependencies:** Full codebase complete.
- **Skills Required:** Docker builds, deployment environments setup.
- **Resources to Learn:** [Dockerizing FastAPI applications](#).
- **Deliverables:** `/deploy/Dockerfile` and `deploy/docker-compose.yml`.
- **Potential Risks:** Oversized Docker images slow to deploy.
- **Acceptance Criteria:** Running `docker-compose up` runs the app and database emulator.
- **Collaboration Required:** DevOps engineer alignment.

Day 29: Handover Documentation Compilation

- **Objective:** Assemble onboarding guides, API docs, and environment instructions.
- **Expected Output:** Clean, versioned documentation repository.
- **Dependencies:** Deployed project structures.
- **Skills Required:** Technical writing, markdown layout.
- **Resources to Learn:** [How to write a developer-friendly README](#).
- **Deliverables:** Final root `README.md` and API specifications PDF.
- **Potential Risks:** Outdated specs mismatching current code.
- **Acceptance Criteria:** A new developer can run the app using README steps in under 15 minutes.

- **Collaboration Required:** Review documents with all team members.

Day 30: Milestone 5 Review: Final Presentation & Code Freeze

- **Objective:** Final presentation and code freeze for Milestone 5 (M5).
- **Expected Output:** Live demonstration to EcoZindagi leads and repository code tag.
- **Dependencies:** Days 26-29 completed.
- **Skills Required:** Public speaking, git branch tag.
- **Resources to Learn:** [Delivering successful software presentations](#).
- **Deliverables:** Tagged release (e.g. `v1.0.0-mvp`) and slide deck.
- **Potential Risks:** Technical hitches during live demo.
- **Acceptance Criteria:** Product Lead accepts project handover.
- **Collaboration Required:** Entire internship cohort (A, B, C, D) and EcoZindagi panel.

3. Gantt-Style Execution Table

Week	Phase / Task Area	Mon (Day 1/6/11/16/21/26/31)	Tue (Day 2/7/12/17/22/27)	Wed (Day 3/8/13/18/23/28)	Thu (Day 4/9/14/19/24/29)	Fri (Day 5/10/15/20/25/30)	Milestones / Checkpoints
W1	Locking & Setup	Project setup & env	ADR comparative analysis	Ingestion schema drafts	Scaffold clean architecture	Scope & ADR lock	M1: Requirements Locked
W2	Ingestion Auth	Device Auth (HMAC)	User Auth (JWT setup)	Telemetry route baseline	Schema Pydantic rules	Mock Ingestion Demo	M2: Ingestion Prototype
W3	Data & Storage	DB connection pool	Telemetry persistence	Latest Status cache	Query logic & page	Access rules & indexes	Database Configured
W4	Rules & Alerts	Engine core setup	Hazard checks rules	Heartbeat monitors	FCM Cloud messaging	Schema freeze meeting	M3: Integration-Ready

W5	Logs & Mocking	Mid-wear log traces	Metrics endpoints	Complete Mock Sim	Pytest execution	Security auditing	M4: Tested & Documented
W6	E2E & Handover	Hardware/Cloud/App test	Validate command checks	Deploy (Docker setup)	Handover docs compile	Final presentation demo	M5: Demo-Ready (Release)